Feeding Asian honeybee queens with European honeybee royal jelly alters body color and expression of related coding and non-coding RNAs

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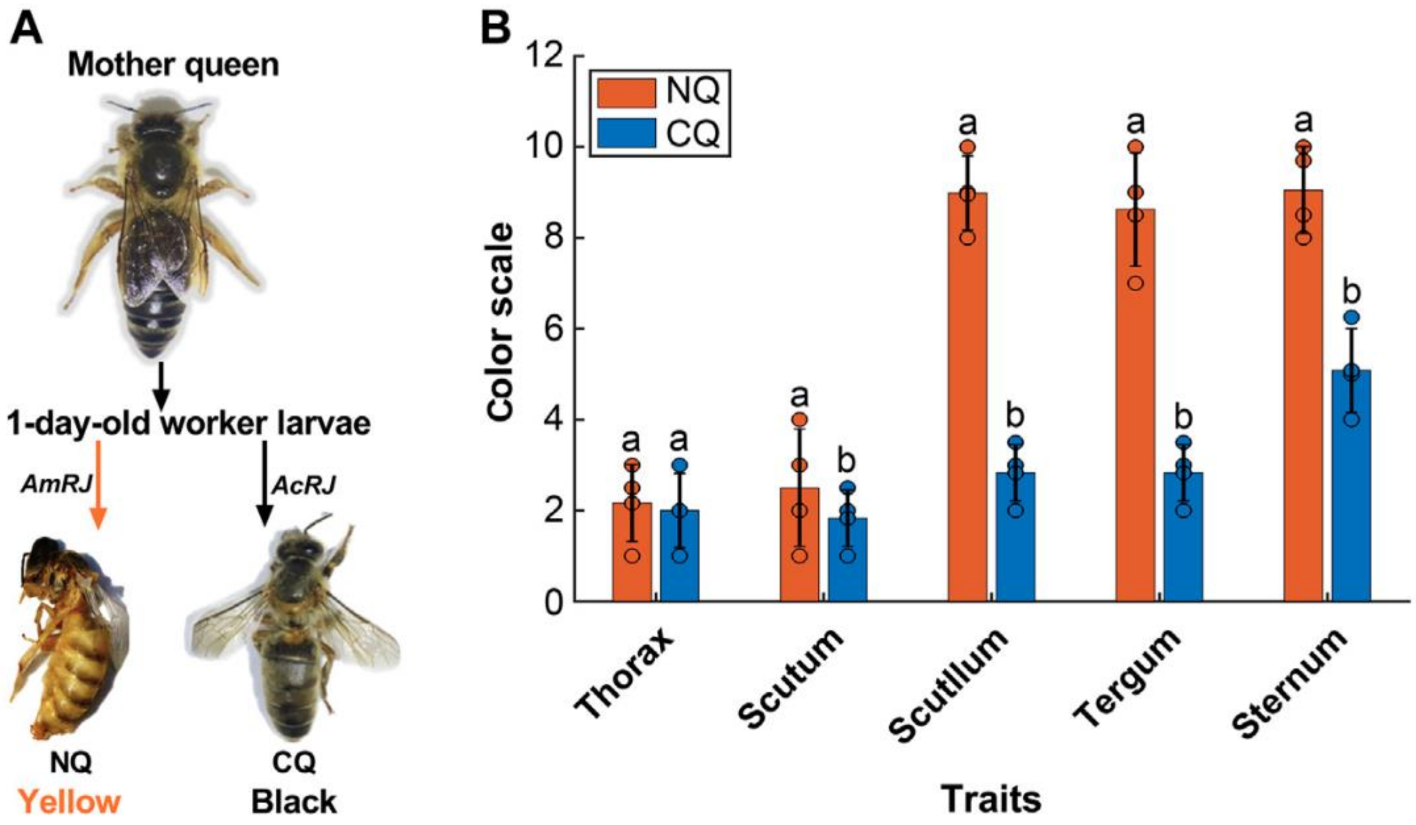
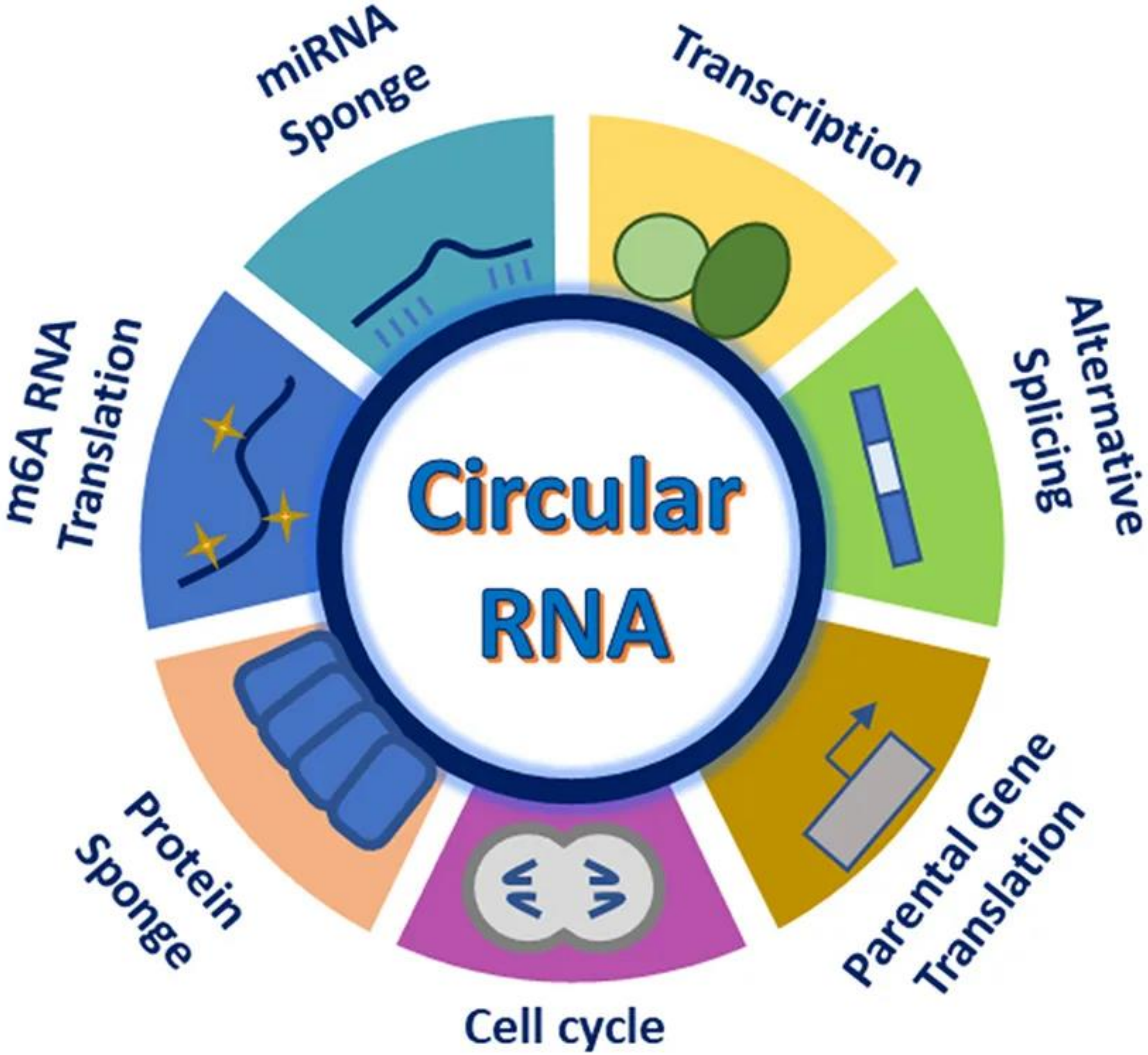
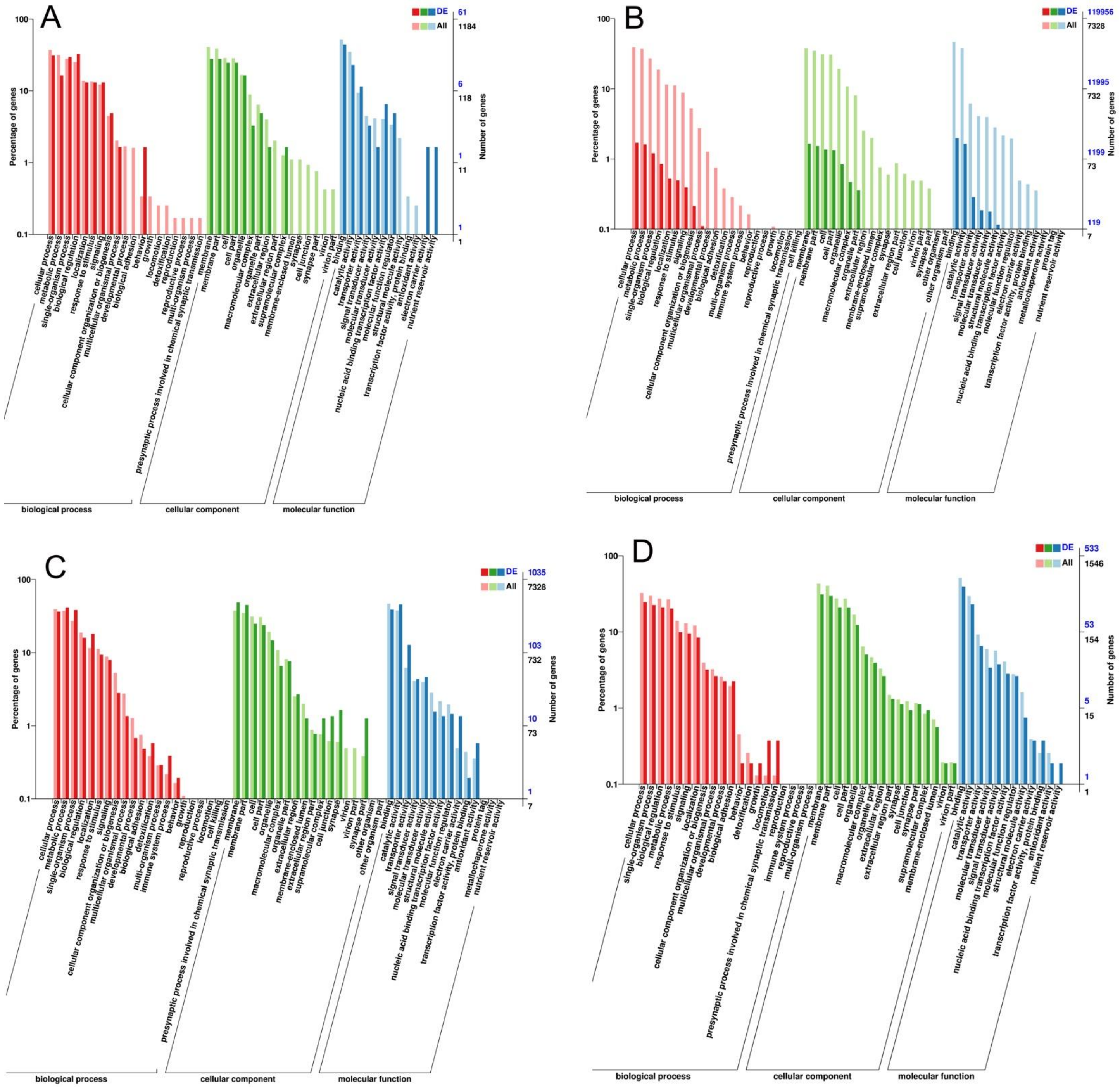


FIGURE 1
(A) The body color alternation in *Apis cerana-Apis mellifera* nutritional crossbreed. The 1-day worker larvae from their *A. cerana* mother queen were fed with *A. mellifera* royal jelly (*Am*RJ) based diet or *A. cerana* royal jelly (*Ac*RJ), resulted in yellow color nutritional crossbreed queens (NQ) or black color control queens (CQ) respectively. **(B)** The body color quantification had significant differences of (NQs, $n = 4$, Scutum's M = 2.5, SD = 1.29, t = 0.85, Scutellum's M = 8.9, SD = 0.81, $t = 5.84$, Tergum's M = 8.5, SD = 1.25, $t = 63.81$, and Sternum's M = 9.5, SD = 0.95, t = 15.4) compared to (CQs, $n = 4$, Scutum's M = 1.83, SD = 0.62 $t = 5.11$, Scutellum's M = 0.81, SD = 0.62, $t = -16.12$, Tergum's M = 2.8, SD = 0.62 $t = 63.81$, and Sternum's M = 5.8, SD = 0.92, $t = 15.14$) all P were < 0.05) based on Ruttner's color scales. Bars present as values of Mean $\pm$ SD, the black dots on the top of each bar represent replicates. Different letters on the top of bars indicate significant difference ($p < 0.05$, Independent-sample t-test).

TABLE 2 Number of differentially expressed coding and non-coding RNAs identified from *A. cerana* artificially nutritional cross queens vs. *A. cerana* natural queens.

NQ vs. *CQ*	DELncRNA		DEGs		DEmiRNA		DEcricRNA	
Regulation	Up	Down	Up	Down	Up	Down	Up	Down
Number of differentially expressed genes or non-coding RNAs	209	102	782	702	45	47	99	70


miRNA Sponge
Transcription
Alternative Splicing
Parental Gene Translation
Cell cycle
Protein Sponge
m6A RNA Translation
Circular RNA



The top enriched and classification of Go terms identified from **(A)** DEcircRNAs, **(B)** DElncRNAs, **(C)** DEGs, and **(D)** DEmiRNAs. The results are summarized in three main categories: biological process, cellular component and molecular function. The lighter colored bars in Y-axis incident the percentages of coding or non-coding RNAs in a category, and the deeper colored ones indicate the present ages of differentially expressed coding or non-coding RNAs.

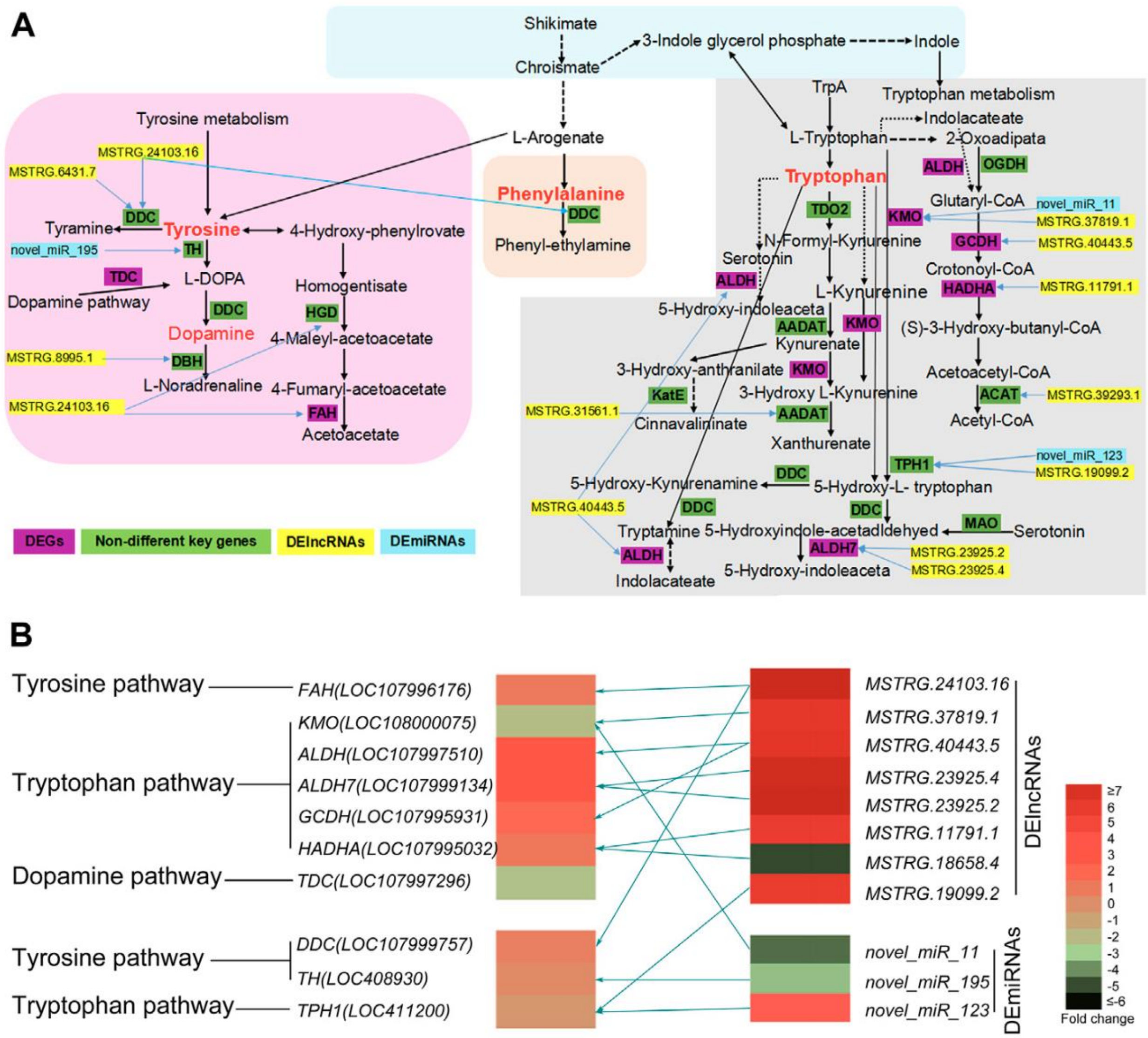


FIGURE 3

(A) The predicted KEGG network of honeybee body color alternation in nutritional crossbreed. This network is based on four key KEGG pathways including tyrosine, tryptophan, dopamine, and phenylalanine pathways. The key DEGs, DElncRNAs and DEmiRNAs involved in this network are also presented. Yellow bars represent DElncRNAs, sky blue bars represent DEmiRNAs, and purple bars represent DEGs and green bars represent key genes but not DEGs. Blue arrows mean DElncRNAs or DEmiRNAs involved into the regulation of related genes. **(B)** The heatmap of key DEGs, DElncRNAs and DEmiRNAs are involved into three key KEGG pathways. Different colors represent significantly differential expression of key DEGs, DElncRNAs and DEmiRNAs using their log2 (Fold change) values. Olive green arrows represent the regulatory relationship between key DEGs and non-coding RNAs (DElncRNAs and DEmiRNAs).

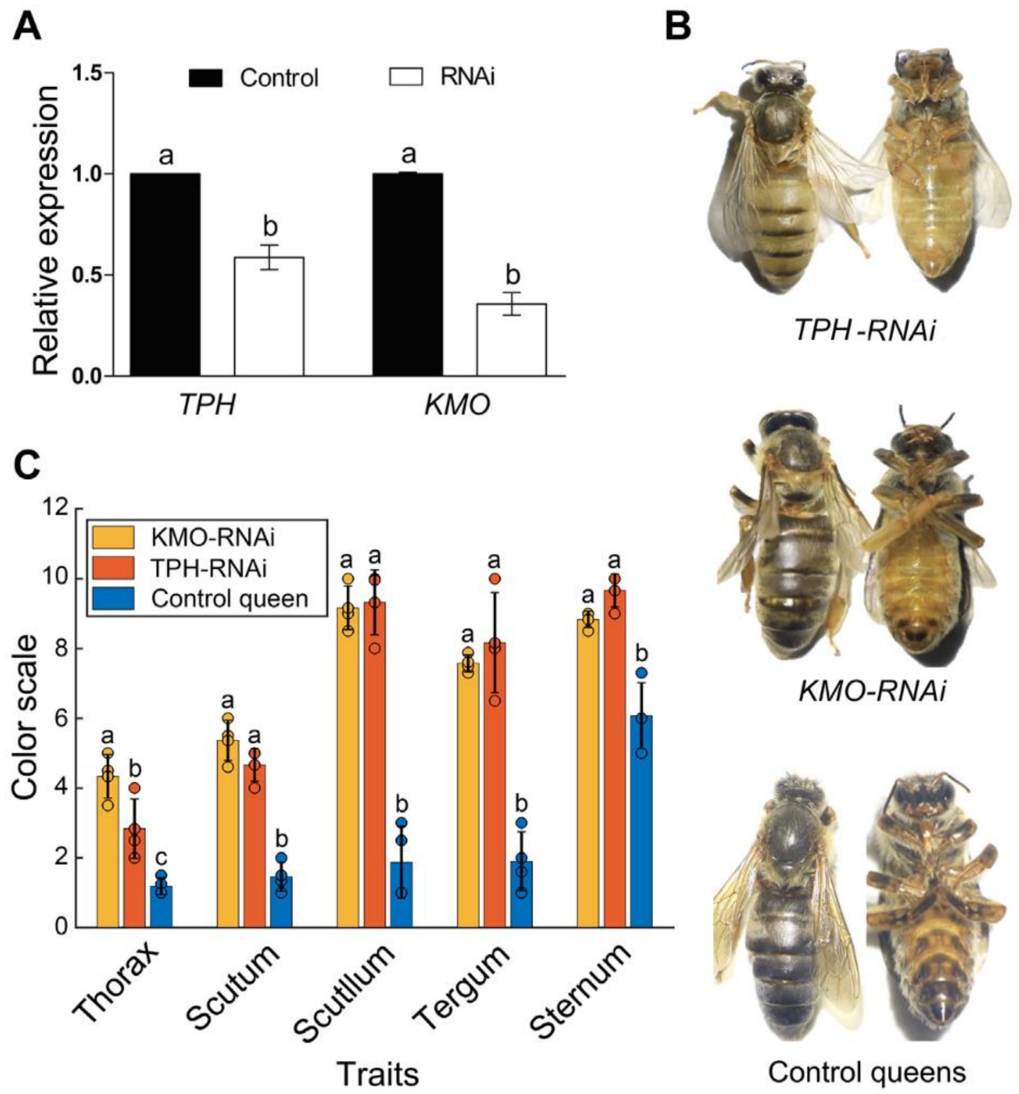


FIGURE 4
(A) The expression of *TPH1* and *KMO* genes in 4-day *A. cerana* queen larvae by RNAi. Bars represent mean ± SD values of relative gene expression. Different letters on the top of each bar represent the significant difference ($p < 0.05$, independent-sample *t*-test). **(B)** The body color alternation of *A. cerana* queens in RNAi experiment. The upper queens with a yellow body color are *TPH1*-RNAi group, the middle ones are *KMO* RNAi group with slight body color changes compared to control bees. The lower ones are control queens fed with negative siRNA. **(C)**The body color quantification of *KMO*-iRNA ($n = 4$), *TPH1*-iRNA ($n = 4$), and control queens ($n = 4$) based on Ruttner's color scales. Bars are present as values of Mean ± SD. Different letters on the top of the bars indicate significant differences ($p < 0.0001$ F = 25.41 F critical = 4.25, One-Way ANOVA), the same letter indicates no significant difference, and the black dots on the top of each bar represent replicates.

“We also identified a **regulation network of eight DElncRNAs and three DEmiRNAs that regulate key DEGs involved in the melanin synthesis pathways**. These results revealed that non-coding RNAs presumably participate in honeybee body color alteration induced by A. cerana-A. mellifera nutritional crossbreeding by regulating the expression of related key genes.”

Review

Epigenetic Mechanisms in *Apis melifera*: From Development to Environmental Adaptation

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Abstract

Epigenetics, as an important scientific field that bridges genomic function and phenotypic plasticity, increasingly demonstrates its value in bee research. In recent years, with the rapid development of omics technologies, there have been significant advancements in the study of epigenetics in honeybees. This article reviews the role of epigenetic regulation in the development, behavioral regulation, and immune response of honeybee larvae from the perspectives of DNA methylation, histone modification, and non-coding RNA. With the continuous deepening of related research, honeybee epigenetics not only opens new paths for understanding the formation mechanisms of complex traits in social insects but also provides solid theoretical support and innovative perspectives for the study of social insects and beekeeping practices. These insights also inform sustainable beekeeping practices.

Overview of Epigenetic Mechanisms in *Apis mellifera*

DNA methylation

- Gene-body & exon enriched; promoter methylation sparse
- Low 5mC (~1% of cytosines)
- Simplified DNMT system: DNMT1 (maintenance) & DNMT3 (de novo)
- Regulates alternative splicing & represses gene expression

Non-coding RNAs

- LncRNAs recruit chromatin modifiers & modulate splicing
- Dynamic expression underlies caste differentiation & neural plasticity
- miRNAs (~22 nt) repress translation via imperfect pairing
- siRNAs (~21–23 nt) cleave mRNA via perfect pairing

Histone modifications

- H3/H4 tails modified by acetyltransferases & methyltransferases
- Activation marks: H3K4me1/2/3, H3K27ac; repression: H3K27me2/3, H3K9me2/3
- Environmental cues (heat stress, spermidine, 10-HDA) reshape marks
- Synergistic with DNA methylation in gene regulation